

1. Project Title & Team

Title: Using the Lottery Ticket Hypothesis to Decrease the Size of LLMs

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2. Problem & Motivation

(About .5 page)

Problem statement: [One crisp paragraph describing the problem you will tackle].

Why it matters: [Who benefits and where it is used (LLM serving, GNN training, recsys, etc.)].

Efficiency demands: [e.g., average latency ≤ 120 ms @ 16K context; VRAM ≤ 40 GB; \$/1M tokens $\leq X$].

Current gap: [1–2 sentences on why existing approaches fall short].

3. Challenges & Trade-offs

(About .5 page)

Bottlenecks: [Compute vs memory vs communication; where is the time/bytes spent?].

Trade-offs: [Latency vs throughput; quality vs latency/energy; accuracy vs model size].

Key measurements to collect: [HBM GB/s, cache hit rate, acceptance rate, accuracies etc.].

4. Proposed Design

(≈ 0.5 page) Core idea: [3–5 bullets of the expected mechanism/system changes].

Why it should work: [Tie to the bottleneck—reduced bytes moved, higher TC util, fewer PCIe trips, etc.].

5. Evaluation Plan

(≈ 0.25 page) Workloads: [Dataset(s), model(s), scale (seq length/graph size), precision].

Environment: [Hardware (A6000/A100/CPU count), platform (Chameleon?), libraries (PyTorch, FAISS, vLLM, DGL/PyG), container/OS].

Metrics: [Serving: TTFT, p50/p95, tokens/s, QPS, VRAM (model+KV), J/token, \$/1M tokens, quality. Training: step time, samples/s, scaling efficiency, HBM/PCIe/NVLink GB/s, energy/epoch, final accuracy. GNN/TGNN: SpMM/SDDMM util, sampler time, feature-store hit rate, end-to-end p95/QPS. RAG: recall@k/nDCG, index p95, embedding/ANN latency, context length].

Baselines: [Dense/stock/paper baseline on your hardware].

Ablations: [Which knobs you will vary: nprobe, PQ code size, LoRA rank, sparsity %, KV precision, etc.].

6. Plan & Milestones

≈0.25 page) Milestone A (Week 6–7): [Baseline + first prototype].

Milestone B (Week 10–11): [Full pipeline + ablations planned].

Final (Week 14–15): [Results + artifacts ready].

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7. References

- [1] Christopher Brix, Parnia Bahar, and Hermann Ney. “Successfully Applying the Stabilized Lottery Ticket Hypothesis to the Transformer Architecture”. In: *CoRR* abs/2005.03454 (2020). arXiv: [2005.03454](https://arxiv.org/abs/2005.03454). URL: <https://arxiv.org/abs/2005.03454>.
- [2] Jonathan Frankle and Michael Carbin. “The Lottery Ticket Hypothesis: Training Pruned Neural Networks”. In: *CoRR* abs/1803.03635 (2018). arXiv: [1803.03635](http://arxiv.org/abs/1803.03635). URL: <http://arxiv.org/abs/1803.03635>.